PHYSICAL CHEMISTRY

Volumetric Analysis

Very Short Answer Question

1. **Deduce the normality equation $N_1V_1 = N_2V_2$. Experimental data obtained by titrating decinormal solution of oxalic acid with potassium permanganate solution are given in the book (p.g 377).**
   a) **Name the above titration.**
   Ans: The above titration is redox titration (oxalic acid acts as a reducing agent and $KMnO_4$ acts as an oxidizing agent).

   b) **Calculate the equivalent weight of $KMnO_4$ from the above data.**
   Ans: Equivalent weight of **$KMnO_4$** is 158 g/mol. Here, n-factor is 5 because when $KMnO_4$ is reduced the change in oxidation number is 5.
   Therefore, Equivalent weight = 158/5 = 31.6 g/equivalent.

   c) **Calculate the normality of $KMnO_4$ form the above data.**
   Ans: Given on 10 set chemistry notebooks.

   d) **Why is dil. H_2SO_4 added to the conical flask containing standard oxalic acid before titrating with $KMnO_4$ solution?**
   Ans: Dilute H_2SO_4 is added to provide an acidic medium because $KMnO_4$ acts as a strong oxidizing agent only in acidic conditions. Without acid, $KMnO_4$ won't fully reduce to Mn^{2+}, and the redox reaction with oxalic acid won't proceed correctly.

   e) **Identify the titrant and titrand in this titration.**
   Ans: Titrant: Potassium permanganate ($KMnO_4$) — taken in the burette.
   Titrand: Oxalic acid solution — taken in the conical flask.

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   a) **Derive the relationship between normality and molarity**
   Ans: Give on notebook.

   b) **Difference between equivalence point and end point.**
   Ans: The point during titration at which completion of reaction is indicated by sharp change in the color of indicator.
   The theoretical end point at which no. of gram equivalent of acid and base exactly reacts with one another to produce neutral solution. There is no change in color.

   c) **If 20 mL of N/2 HCl, 60 mL of N/10 HNO_3 and 150 mL of N/5 H_2SO_4 mixed together, what will be the normality of the resulting acid mixture?**
   Ans: Given on 10 set chemistry notebooks.

3. **Titration is a common volumetric method for the determination of strength of unknown solution. Two major components of titration are titrand and titrant.**

   a) **Distinguish between titrand and titrant.**

   Ans: Titrant is the solution of known concentration added from the burette during titration.

   Titrand is the solution of unknown concentration present in the conical flask that reacts with the titrant.

   b) **Write two major requisites for a substance to become primary standard.**

   Ans: Two major requisites for a substance to become primary standard are

   - The substance must be highly pure, stable and non-hygroscopic
   - It should be stable even in solution form, even on standing for a long time.

   Note: ◆ Hygroscopic

   A substance that absorbs moisture from the air but does not dissolve in it.

   Example: Concentrated sulfuric acid, NaOH pellets.

   📘 Definition: Hygroscopic substances attract and hold water molecules from the surrounding air.

   ◆ Efflorescent

   A substance that loses water to the air when exposed, appearing dry or powdery.

   Example: Washing soda (Na2CO3·10H2O)

   📘 Definition: Efflorescent substances lose water of crystallization to the atmosphere.

   ◆ Deliquescent

   A substance that absorbs so much moisture from the air that it dissolves in it and forms a solution.

   Example: Calcium chloride ($CaCl_2$)

   📘 Definition: Deliquescent substances absorb atmospheric moisture until they dissolve completely.

   c) **Calculate the molality of salt (molar mass = 138g/mol) solution obtained by dissolving 2.5 g of salt of salt in one litre of its solution. Given that density of the solution is 0.85 g/cm^3**

   Ans: Given on 10 set chemistry notebooks.

   d) **Deduce the relation: $N_1V_1 = N_2V_2$**

   Ans: Given on 10 set chemistry notebooks.

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   a) **Give reason:**

   - **Methyl orange is used as indicator for the titration between H_2SO_4 and Na_2CO_3.**

     Ans: Methyl orange is suitable because the reaction between H_2SO_4 (a strong acid) and Na_2CO_3 (a weak base) ends in an acidic medium, and

methyl orange changes color in the pH range 3.1–4.4, which lies in the acidic zone.

- **Oxalic acid solution is warmed by adding dil. H_2SO_4 before titration with $KMnO_4$.**
  Ans: Oxalic acid is warmed (around 60–70°C) after adding dilute H_2SO_4 to increase the rate of reaction, because at room temperature the reaction with $KMnO_4$ is very slow. Heating helps oxalic acid decompose faster, allowing smooth and complete redox titration.

b) **0.70 g of a sample $Na_2CO_3 . xH_2O$ was dissolved in water and volume was made 100mL. 20 mL of this solution required 19.8 mL in N/10 HCl for complete neutralization. Find the value of x.**
Ans: Given on 10 set chemistry notebooks.

5. **The concentration of H_2SO_4 solution can be determined by the titration with standard Na_2CO_3 solution.**
   a) **Is the Na_2CO_3 solution primary standard? Why?**
   Ans: Yes, Na_2CO_3 is a primary standard because it is stable, pure, non-hygroscopic, and has a high molar mass, making it easy to weigh accurately and prepare a solution of known concentration.

   b) **100mL of Na_2CO_3 solution contains 0.53g of Na_2CO_3. If 10mL of this solution is added to "x" mL of water to obtain 0.01 M Na_2CO_3 solution. Calculate the value of x.**
   Ans: Given on 10 set chemistry notebooks.

6. **Normality factor and titration error are important in analytical chemistry for the determination of variation of the observed result for the same experiment.**
   a) **Distinguish between normality factor and titration error.**
   Ans: Normality factor is the ratio of the actual (experimental) normality of a solution to its theoretical (expected) normality. It is used to correct the concentration of a solution in titrations.
   Titration error is the difference between the end point and the true equivalence point of a titration. It arises due to indicator lag or improper technique.

   b) **A chemistry lab-boy weighed 1.31 g of Na_2CO_3 to prepare 250 mL of its decinormal solution.**
      - **Calculate the actual strength of the solution indicating normality factor.**
        Ans: Given on 10 set chemistry notebooks.
      - **Find the volume of water required to be evaporated to make it exactly normal solution.**
        Ans: Given on 10 set chemistry notebooks.
      - **Calculate the mass of oxalic acid crystal required to neutralization of solution completely.**
        Ans: Given on 10 set chemistry notebooks.

c) **How would you apply of titration in pharmaceutical (medicines and the drug industry) industry?**
Ans: Titration is used in the pharmaceutical industry to determine the exact concentration and purity of drugs and active ingredients. It ensures correct dosage formulation and checks for contaminants or impurities during quality control.

7. **End point and equivalent point are the common terminologies used in titration.**
   a) **What is meant by titration error and how can it be minimized?**
   Ans: Titration error is the difference between the end point (color change) and the equivalence point (actual neutralization).
   It can be minimized by using a suitable indicator, adding titrant drop by drop near the end, and using standard solutions.

8. **Normality and molarity are the terminologies of volumetric analysis used to express the concentration of solution.**
   a) **Distinguish between molarity and normality**
   Ans: Molarity (M) is the number of moles of solute per liter of solution.
   Normality (N) is the number of gram equivalents of solute per liter of solution.

   b) **Normality equation derivation**
   Ans: Given on copy.

   c) **Why is this equation not always used to calculate molarity?**
   Ans: The normality equation N1V1=N2V2N is not always used to calculate molarity because normality depends on the reaction type (acid-base, redox) and the number of equivalents, which can vary, while molarity is simply moles per liter and does not account for equivalence factors. So, normality changes with the nature of the reaction, making the equation unsuitable for direct molarity calculations in all cases.

   d) **Your chemistry teacher added 4 g of sodium hydroxide in a bottle containing 20 cc of 2N H_2SO_4 and he diluted it up to 1 liter by adding water. Then, he gave you a blue and a red litmus paper.**
      - **Which litmus paper would you used to test the solution and why?**
        Ans: Given on NB note copy.
      - **Calculate the normality of the dilute solution.**
        Ans: Given on NB note copy.

9. **Concentration of sulphuric acid solution can be determined with titration with primary standard sodium carbonate solution.**
   a) **Na_2CO_3 is considered as primary standard why?**
   Ans: Na_2CO_3 is considered as primary standard because it is stable, pure, non-hygroscopic, and has a high molar mass, making it easy to weigh accurately and prepare a solution of known concentration.

b) Which chemical indicator is suitable for this titration?

Ans: Methyl orange is ideal for strong acid (H_2SO_4) vs weak base (Na_2CO_3) titrations because it changes color in the acidic pH range (3.1–4.4), signaling the end point clearly.